

Joint scheduling and power control

- Exploiting scalable video coding
- Modeled as a multiple-choice knapsack problem
- Solution uses FPTAS and dynamic programming

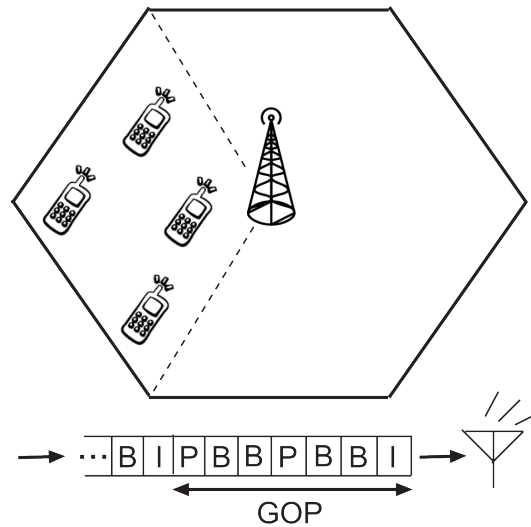


Fig 1. Each user in the same sector transmits a video to a common base station. Frames arrive one-by-one, and are stored in a buffer before being transmitted.

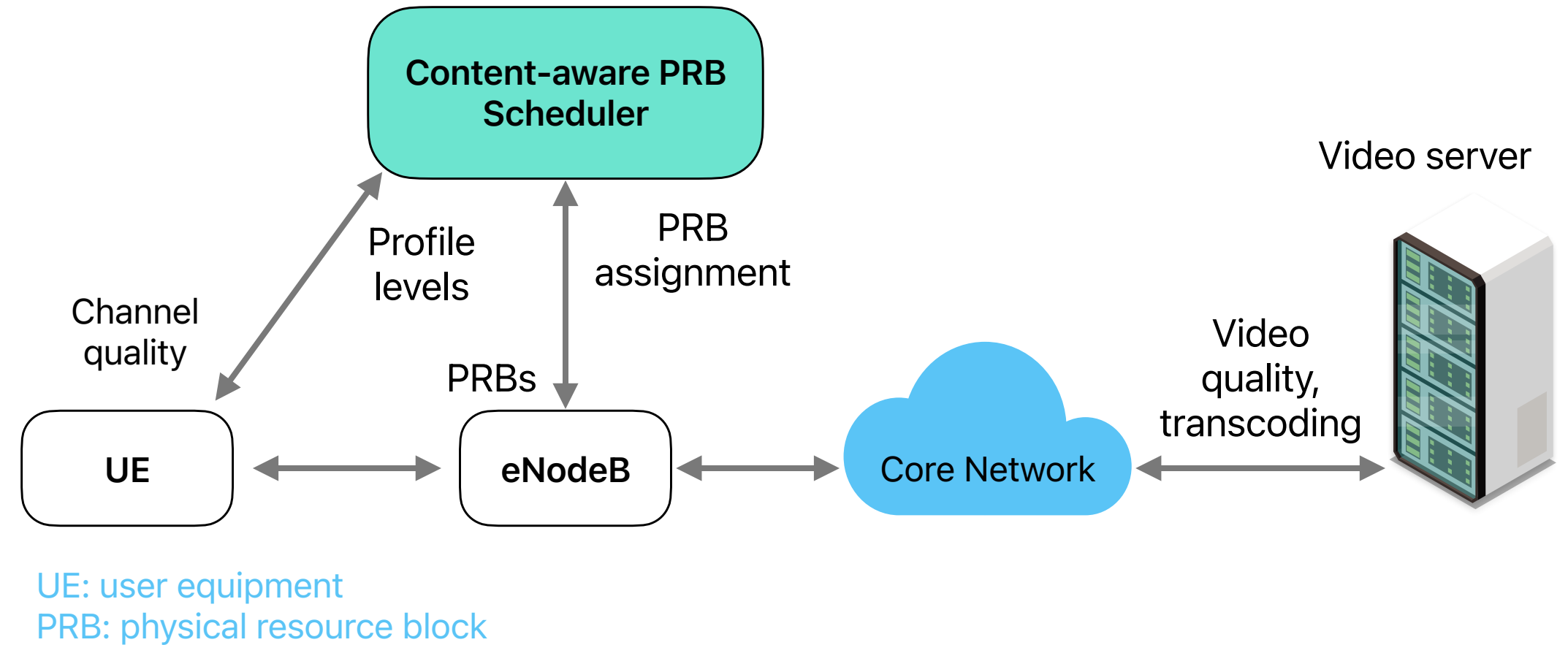


Fig 2. A content-aware architecture for video delivery over LTE downlink. Solid lines indicate interfaces that already exist in an LTE system. Dotted lines are the new conceptual interfaces proposed to implement the PRB scheduler.